

Varsha Giridharan

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OBJECTIVE: Working with AI and ML tools to solve real-world problems.

EDUCATION

MS in Applied Data Analytics

Boston University

CGPA – 3.57 / 4.00

Jan 2025*

B Tech in Computer Science and Engineering with Specialization in AI and ML

SRM Institute of Science and Technology, Chennai

CGPA – 8.81/10.00

May 2023

12th Grade, Padma Seshadri Bala Bhavan Senior Secondary School, Chennai

76.4%

May 2019

10th Grade, Padma Seshadri Bala Bhavan Senior Secondary School, Chennai

70.3%

Jun 2017

RESEARCH INTERESTS: ML, Data Science and Analytics, AI

COURSEWORK TAKEN: Data Structures and Algorithms, Natural Language Processing, AI, Applied Machine Learning

WORK EXPERIENCE

Sept 2023 - Ongoing

Research Assistant, Boston University

- Web scrape all possible user-to-user interactions from the DeviantArt website and store all the data in a database.
- Learning more about Neural Networks
- Working with Classification and Regression
- Working with Keras
- Learning about Deep Learning for Computer Vision
- Working with Large Language Models

INTERSHIPS

Data Scientist, Let's Grow More

Mar 2022 – Apr 2022

-a not-for-profit E-learning platform focusing on tech-based upskilling, with 1089 employees.

- Solved a given set of problem statements within a month by watching 10 pre-recorded instructional videos.
- Gained hands-on experience in ML, model creation, model prediction, data analysis, and data visualization by working on 5 assigned projects such as Stock Market Prediction and Forecasting, and Image to Pencil Sketch with Python.

RESEARCH EXPERIENCE

Design and Development of Smart Farming using ML and IoT in India (Team Size – 3)

Jan 2023 – May 2023

- The goal of this research work is to deploy most ML and IoT solutions to help farmers with soil analysis and yield forecasting.
- Machine Learning predicts, the yields which require specific Indian climate conditions using techniques such as K-mean clustering with the Elbow method, logistic regression, and learn and test values.
- The IoT gadget called REES52 includes sensors like the soil moisture sensor and a simple and incredibly affordable digital temperature and humidity sensor. Measures ambient air using a thermistor and a capacitive humidity sensor.

UNDERGRADUATE PROJECT

Object Detection on NVIDIA Jetson Nano (Minor project, Team Size – 3)

Aug 2022 – Nov 2022

- Built an object detector using YOLOv5 and OpenCV on NVIDIA Jetson Nano.
- Created a pipeline using CSI Camera; Detected multiple objects using IMX477-160 12.3 MP CSI camera and YOLOv5 model and implemented the YOLOv5 network model in the PyTorch framework.
- Captured image with 12 frames per second; Detected and labelled multiple objects with an F1 score of 1.0.

COURSE PROJECTS

Sentiment Analysis

Mar 2022 – Apr 2022

- Designed a sentiment analysis tool in Google Colab to determine the sentiments and opinions behind tweets.
- Built 4 models using linear regression, logistic regression, random forest classifier, linear SVC and multinomial Naïve Bayes algorithms; trained these using data converted from pre-labelled tweets in the Sentiment140 database.
- Tested the algorithms using data from a test dataset and compared the predictions to the right answers using a confusion matrix. Recorded an accuracy of 70-75% for all the 4 models, despite variations in training time.

Fake News Detection

May 2022 – Jun 2022

- Developed a deep learning model to detect whether a given news is fact or fake.
- Built an MVP with the BERT stand-alone model and implemented the BERT TF 2.0 algorithm using TensorFlow 2.0 for classification; trained the model using GPU available in Google Colab.
- Achieved an accuracy of ~95% on test set, and an AUC score of 0.99285 with the standalone BERT Model.

Speech Emotion Recognition

Jun 2022 – Jul 2022

- Designed a ML model to recognize 4 distinct emotions – happiness, sadness, anxiety, and depression – from speech.
- Built the audio-processing model using Python and ML libraries like librosa, soundfile and sklearn. Split data from RAVDESS dataset in JupyterLab into train and test datasets. Classified emotions from sound files using MLPClassifier.
- Recognized human emotion and affective states from speech with an accuracy of 72.4%.

Gender and Age Detection

Jul 2022 – Aug 2022

- Designed a ML model that can predict the gender and age of a person using Google Colab.

- Built the model using Python and ML libraries like cv2, cv2_imshow, math and time. Detected faces and classified them based on gender and age using OpenCV DNN module which supports DarkNet framework.
- Predicted the age and gender with an accuracy of 100%

Credit Card Fraud Detection

Sept 2022 – Oct 2022

- Developed a ML model to eliminate fraudulent credit card transactions using Google Colab.
- Built the model using Python and ML libraries; Developed a classifier to detect fraudulent transactions with minimal delay using Logistic Regression, SVC, KNeighborsClassifier and Random Forest Classifier algorithms.
- Recorded a final AUC score of 0.794737 in the logistic regression test set using the Undersampling technique.

CONFERENCES ATTENDED

May 2023

- 11th International Conference on Contemporary Engineering and Technology

PUBLICATIONS

(yet to be published)

- Design and Development of Smart Farming using ML and IoT in India, Indonesian Journal of Science and Technology (Scopus).

CERTIFICATIONS

- Certificate of Presentation in the 11th International Conference on Contemporary Engineering and Technology, Organization of Science and Innovation Engineering and Technology (OSIET) affiliated to Anna University, May 2023
- The Fundamentals of Digital Marketing, Google, Aug 2022
- Getting Started with AI on Jetson Nano, NVIDIA Deep Learning Institute, Aug 2022
- Building Video AI Applications at the Edge on Jetson Nano, NVIDIA Deep Learning Institute, Aug 2022
- Data Structures & Algorithms in Python, Coding Ninjas, Jul 2022
- Database Foundations, Oracle, Oracle Academy, Apr 2022
- UI/UX & Graphic Designing, Tech Analogy, Apr 2022
- Artificial Intelligence and High-Performance Computing, Intel Software, Fice, Nov 2021
- Introduction to Data Science, IBM, Nov 2021
- Data Science Methodology, IBM, Nov 2021
- Python 101 for Data Science, IBM, Nov 2021
- Open Data Foundations, IBM, Nov 2021
- Data Science with Scala, IBM, Nov 2021

TECHNICAL SKILLS: Microsoft, Linux, C Programming, Object Oriented Programming Language (C++), Python3, Java, MySQL, HTML, Java Database Connectivity (JDBC), Visual Studio Code, Google Colab, Github, , Adobe Illustrator, Adobe Photoshop, Star UML, Jupyter Lab, Jupyter Notebook, Spyder, Datalore, IBM Watson Studio Cloud, Glueviz, Orange3, PyCharm, RStudio, Data Spell

CO-CURRICULAR AND EXTRA-CURRICULAR ACTIVITIES

- *Chess player*, Chess Club, SRM Institute of Science and Technology, participated in daily chess practice matches for 45 minutes with 100+ student members, Jun 2019 – Present.
- *Member*, White Hat Hackers Club, SRM Institute of Science and Technology: assisted in managing the registrations of hackathon participants, Jun 2019 – Present.
- UX/UI Design Scape Workshop (45 days), Tech Analogy: designed a dating venue finder application in a competition (200+ participants), and received invite to join the Tech Analogy community (4811 members), Mar 2022 – Apr 2022.
- Artificial Intelligence Workshop (1 day), Design and Innovation Club, SRM Institute of Science and Technology: attended talks on the real-life applications of AI and Data Science, Jan 2022.
- *Volunteer*, National Cadet Corps, SRM Institute of Science and Technology: participated in a reforestation awareness march in Vadapalani along with 58 students, and exhibited cooking skills at a skill workshop, Jun 2019 – Nov 2019.

LANGUAGES: English, Hindi, Tamil (Read, Write, Speak), French (A1).